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Publications and Preprints

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Published and Accepted Papers

[1] Xiang-dong Hou and **Shujun Zhao**. On a conjecture about the sum-freedom of the binary multiplicative inverse function. *Designs, Codes and Cryptography* **94** (2026), Article 86. doi:10.1007/s10623-026-01819-9.

[2] Alyssa Ebeling, Xiang-dong Hou, Ashley Rydell, and **Shujun Zhao**. On sum-free functions. *Finite Fields and Their Applications* **110** (2026), Article 102744. doi:10.1016/j.ffa.2025.102744.

[3] Xiang-dong Hou and **Shujun Zhao**. Two absolutely irreducible polynomials over $\mathbb{F}_2$ and their applications to a conjecture by Carlet. *Finite Fields and Their Applications* **109** (2026), Article 102713. doi:10.1016/j.ffa.2025.102713.

[4] Luca Bastioni, Giacomo Micheli, and **Shujun Zhao**. On the characteristic polynomial of linearized polynomials. *Mathematics of Computation*, to appear. arXiv:2506.16937.

[5] Mohamed O. Darwish, Anuvrat Jaindungarwal, Giacomo Micheli, Divyesh Vaghasiya, and **Shujun Zhao**. Good polynomials for locally recoverable codes: Classification results, asymptotics, and exact formulae. *Journal of Algebra and Its Applications* (2025), Article 2541017. doi:10.1142/S0219498825410178.

Preprints

[6] Xiang-dong Hou and **Shujun Zhao**. Further results on sum-freedom of binary and q-ary functions. Preprint (2026). arXiv:2609.31489.

[7] Xiang-dong Hou, Siyu Peng, Yongyu Qiang, and **Shujun Zhao**. Classification of rational functions of degree three over finite fields. Preprint (2026). arXiv:2604.18954.